

Rate My Course

Final Presentation

Agile Development Processes (EDA397)
Group 10 | 13.03.2025





Table of contents

01

**Application
Overview**

02

**Development
Process**

03

**Project
Assessment**

04

Next Steps

05

**Team
Collaboration**

06

**Question
Round**



01

Application Overview





Introduction

Problem at Universities



Student

Lack of Information for Course Selection

University

Limited Feedback for improvement insights

Our Product Vision



Improved Decision-making During Course Selection

Providing Valuable Feedback to Improve Course Quality



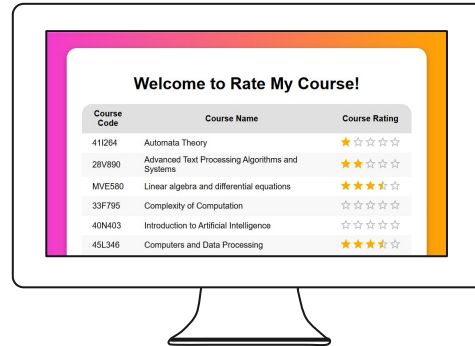
Key Features

Browse & Search

Students can browse and search courses

Rate & Review

Students can provide feedback on courses



Admin Moderation

All reviews undergo manual approval before being displayed

User Management

Registration of student accounts, login, and verification

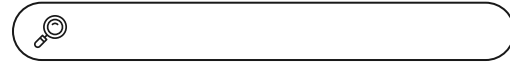
Profanity Filtering

Automatic detection of profanity in reviews

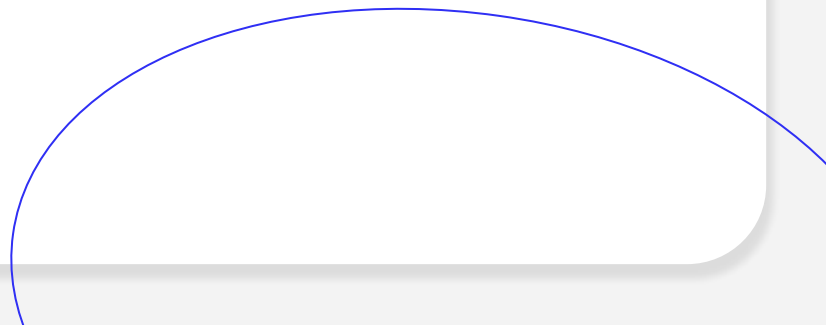
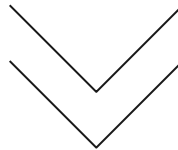
Commenting

Students can leave comments on other reviews

⚡ Changes due to Customer Requests



Live Demonstration





02

Development Process





Sprint Increments

Sprint 1

Initial Functionality for User Access & Navigation

- Account Management with User registration and sign-in
- Enabled browsing of courses and reviews

Sprint 2

Building the Review System & Moderation

- Students can post their reviews
- Administrators can manually approve reviews

Sprint 3

Enhancing User Engagement & Moderation

- Students can like & dislike reviews
- Students can comment on reviews
- Implemented automatic profanity filtering



Supporting Tools



Communication Channel



Shared Code Repository



Sprint Planning &
Progress Tracking



Internal Documentation





Role Assignment

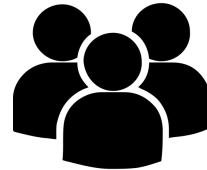
Rotation of Roles at the Beginning of each Sprint



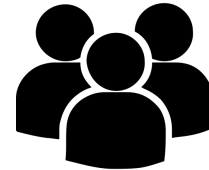
Product Owner



Scrum Master



Frontend Team



Backend Team



Product Backlog

Backlog Updates across Sprints:

- Started with 10 user stories.
- Added 5 new user stories
- Completed 9 user stories

Backlog Refinement Sessions:

- Held twice per sprint
- Adapt to customer needs and new insights
- Clarification and discussion of items



Projects / EDA 397

Backlog

Q Search



Epic ▾

Type ▾

☐ > SCRUM Sprint 3 27 Feb – 13 Mar (23 issues)

☐ ▾ Backlog (6 issues)

SCRUM-2 As a user, I want to be able to change my password so that it is easier to reme...

SCRUM-3 As a user, I want to be able to reset my password so that I can log in to my acc...

SCRUM-4 As a user, I want to be able to delete my account so that my data is removed fr...

SCRUM-5 As a user, I want to be able to unsubscribe from mailing lists so that I am no lon...

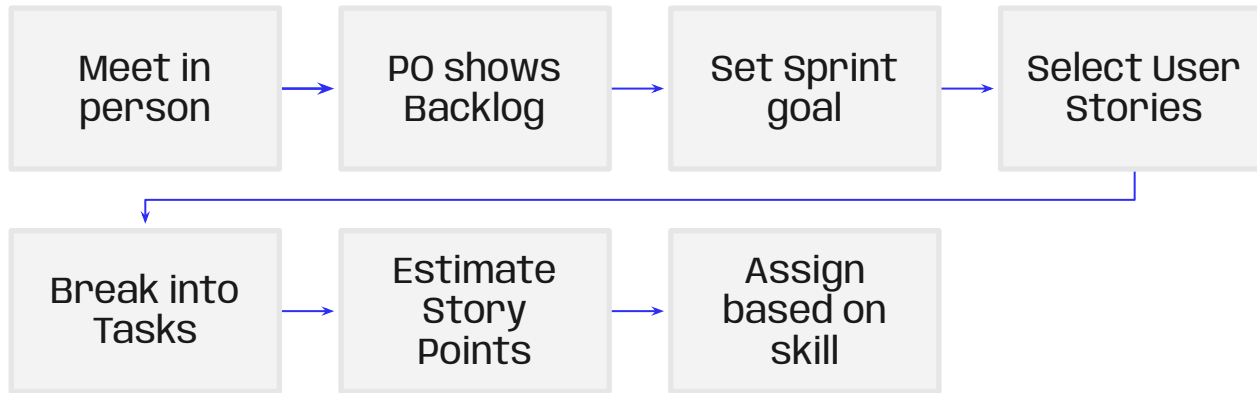
SCRUM-7 As an administrator, I want to be able to register professor and course data ont...

SCRUM-66 As a user, I want to be able to report a review comment, so as to ensure offens...

+ Create issue



Sprint Planning



Challenges:

- Breaking Down Tasks to Be Independent
- Different Skill Sets Within the Team
- Varying Availability of Team Members



Sprint Backlog

☐ **SCRUM Sprint 3** 27 Feb – 13 Mar (23 issues)

2 5 23

Complete sprint



- | | | | | |
|--|--|--------|---|----|
| ☒ SCRUM-69 | Implement UI to display replies for each review | DONE ▾ | 4 | JH |
| ☐ SCRUM-57 | Course page crashes if a course has no professors assigned | DONE ▾ | 0 | JH |
| ☐ SCRUM-68 | Fix course page pagination | DONE ▾ | 0 | JH |
| ☒ SCRUM-58 | Populate the database for future demonstrations | DONE ▾ | 1 | ML |
| ☒ SCRUM-60 | Integrate a profanity filter onto the backend API | DONE ▾ | 2 | ML |
| ☒ SCRUM-70 | Create database representation for review reactions | DONE ▾ | 1 | ML |
| ☒ SCRUM-72 | Add likeCount and dislikeCount values to the review resource | DONE ▾ | 1 | ML |
| ☒ SCRUM-71 | Create endpoints for posting reactions | DONE ▾ | 1 | ML |



Sprint Execution

Tri-Weekly Standups



Started with dailys but changed the schedule due to availabilities and lack of updates

Coding Approach



Tried pair programming but mostly worked individually based on personal preference.

Scope Changes



Occurred as tasks became clearer, bugs and dependencies, surfaced and missing tasks were identified.

Peer Reviews



Peer Reviews via Pull Requests with assigned reviewer pairs

Documentation in Confluence



Documentation of scope changes, retrospectives, API endpoints, and database schema was maintained for smooth handovers.



Documentation

Organizational



Sprint Backlog - Changes



Owned by [Marleen Lukei](#) · · · ·

Last updated: just a moment ago · [👤](#) See how many people viewed this



This page is used to document changes made to the sprint backlog during the sprint.

Story Points	Identifier	Reason	Date
4	SCRUM-45 and SCRUM-50	Separate Creation of User Pages (Admin and Student)	21.02.2025
1	SCRUM-52	Add missing course descriptions	25.01.2025
1	SCRUM-47	Modify Home Page after the Login	18.01.2025
Bug	SCRUM-46	Professor search by course numbers	18.02.2025

Technical

Compilation Details



Owned by [Lucas Lonegro Gurfinkel](#) · · · ·

Last updated: Feb 11, 2025 by [Jonas Chua Shan Hong](#) · 1 min read · [👤](#) See how many people viewed this

Compiling is done in the backend folder with:

```
mvn clean package
```

When the application is initialized, database scripts run to create the necessary relationships.

The following environment files must be created in the `/webapp/src/main/resources` directory:

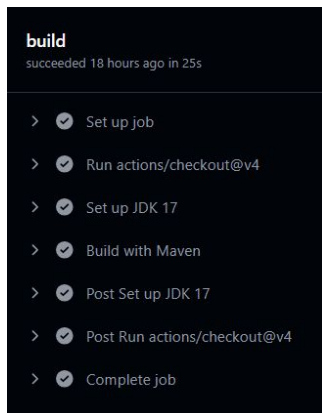
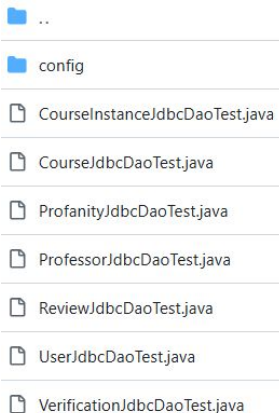
- `application.properties` containing:
 - `postgres.user` - username for the postgres database
 - `postgres.password` - password for the postgres database
 - `databaseUrl` - url for the postgres database
 - `maxFileSize` - maximum size of an image upload that will be accepted by the backend server
- `rememberme.key` - containing a secret key for token based remember me authentication. An appropriate way of generating this key is `openssl rand -base64 2048 > rememberme.key`
- `jwt.key` - containing a secret key for jwt authentication. An appropriate way of generating this key is `ssh-keygen -t rsa -b 4096 -m PEM -f jwt.key`



Testing

```
@Test
public void testFindById() {
    Optional<CourseInstance> cou

    Assert.assertNotNull(courseI
    Assert.assertTrue(courseInst
    Assert.assertEquals(PREEXIST
}
```



JUnit tests in the backend with automated build and test execution on each push to the main branch



Testing

Manual Testing Document For The Frontend Application



Testing Documentation



Owned by [David Viskovic](#) ...

Last updated: Feb 27, 2025 • 2 min read • [Let](#) See how many people viewed this

By following the steps below and verifying the expected results, the key functionalities of the frontend application can be considered as functional.

Test Case: Course Page Access

Steps:

1. Navigate to the home page (/).
2. Click on a course from the list to navigate to the course page.

Expected Result:

- The course page should display the course details, including description, professors, and average rating.
- The reviews section should display a list of reviews with sorting options.

Test Case: Login Functionality

Steps:

1. Navigate to the login page (/login).
2. Enter valid user credentials in the username and password fields.
3. Press the "Login" button.

Expected Result:

- A loading visual (spinner) should appear.
- After a short delay, the user should be redirected to the home page (/).
- The navigation bar should display the "Profile" and "Logout" buttons, indicating that the user is authenticated.



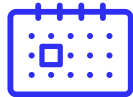
Sprint Monitoring



Customer Feedback



Sprint Retrospective



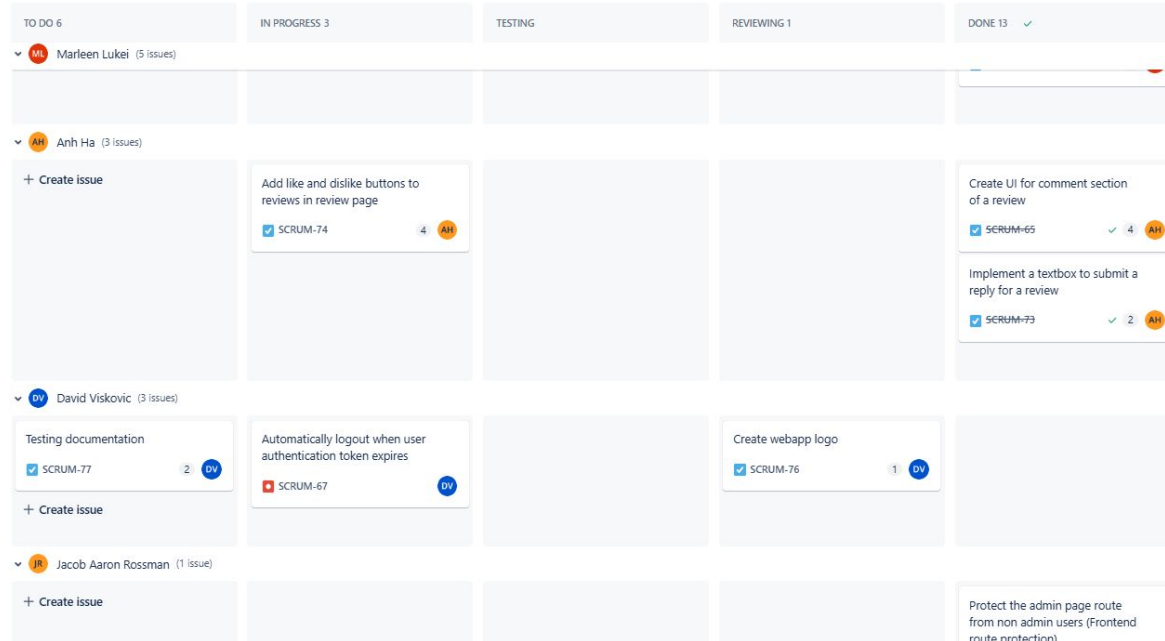
Sprint Board



Burndown Charts



Sprint Board



Provides insights on an individual team member:

- Who is responsible for which task
- Signs of uneven workload distribution
- Shows where progress is blocked or delayed



Sprint Retrospective

Sprint Retrospective



Owned by [Anh Ha](#), created with a template ...

Last updated: Feb 13, 2025 by [Marleen Luke](#)  See how many people viewed this



At the end of each sprint, we conduct a sprint review to evaluate our work during the previous sprint, and our overall progress towards our objectives.

Sprint 1



Sprint Objective: (fill in after we define our sprint objectives)

What went well?	What did not go well?	What have you learned?	What could be done better?
Communication/ sticking to meeting times	Unidentified Dependencies between frontend and backend lead to waiting on each other to finish tasks Task estimations	Scale task estimations more sparsely Split task more granularly	Need two repositories for backend and frontend



03

Project Assessment





Team Performance

Metric	Sprint 1	Sprint 2	Sprint 3	Trend
Sprint Velocity	21 Story Points	25 Story Points	32 Story Points	~ 26 Story Points ↗
Sprint Completion Rate (Done/All SP)	21/30 (70%)	25/27 (92.6%)	32/32 (100%)	~ 87.5% ↗
Average Time until set to Done	8.5 Days	11 Days	7.5 Days	~ 9 Days ↘
Average Time in Status Review	?	3.9 Days	1.4 Days	~ 2.65 Days ↘
Sprint Scope changes	?	+ 6 Story Points + 2 Bug Fixes	+ 4 Story Points + 1 Bug Fix	~ + 5 Story Points ↘ ~ + 1-2 Bug Fixes ↘

*?: indicates lack of reliable data since in the first sprint this was not documented



Team Performance

Burndown Charts per Sprint



Sprint 1



Sprint 2



Sprint 3



Product Delivered

Planned Requirements	Status	Comment
Browse and search reviews	✓	
Rate and review courses	✓	
Account management with registration and login	✓	
Manually approve course reviews as an administrator	✓	
Comment on reviews	↻	Scope Changes conducted excluding editing and reporting of comments
Like or dislike comments	✓	
Automatic filtering using a profanity filter for reviews, comments and usernames	✓	
Enhanced account management: change password, reset password, delete account	✗	Postponed due to low priority
Register professors and courses as an administrator	✗	Postponed due to low priority

* Blue boxes represent customer requests



04

Next steps





Deployment Strategy



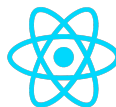
PostgreSQL



Microsoft
Azure



JUnit
spring®



cyPress
React

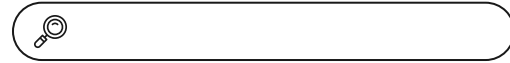




05

Individual Reflections





Individual Reflection on...



Roles assumed
in each sprint



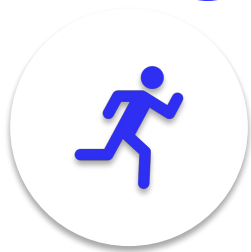
Contribution to
the product



Key Takeaways



Individual Reflection for Zheng Wei



Roles assumed
in each sprint

- Sprint 1: Backend Developer
- Sprint 2: Frontend Developer
- Sprint 3: Product Owner with Anh, Frontend Developer



Individual Reflection for Zheng Wei



Contribution to
the product

- Sprint 1: Technical documentation
- Sprint 2: Profile Page
- Sprint 3: Approve /Reject functionality for reviews



Individual Reflection for Zheng Wei



Key Takeaway:
Continuous
Integration

- Initially: only main branch
- vagueness of project programming languages
- started separately & not integrated frequently
- discard waste and favour 1 path
- Now: different branches



Individual Reflection for Marleen



Roles assumed in
each sprint

- Sprint 1: Backend Developer
- Sprint 2: Scrum Master and Backend Developer
- Sprint 3: Backend Developer



Individual Reflection for Marleen



Contribution to
the product

- Sprint 1: Database Setup & Seeding
- Sprint 2: Approved Review Filter for Students, Extended Course Endpoint
- Sprint 3: Profanity Filtering, Reaction to Reviews (Like/Dislike)

Overall Responsible for Course Deliverables:
Collecting Feedback to incorporate into Sprint
Reports and Group Presentation



Individual Reflection for Marleen



Key Takeaway:
Clear Processes
and Accountability

Initial Perspective: If task is put into to the review column, just mentioning it should be enough and “someone” will feel responsible

Problem: Task remained stuck in review stage, bottleneck as Queue began to form

Actions: explicitly assign reviewers and have them move the items into “Done”



Individual Reflection for Jacob



Roles assumed in
each sprint

- Sprint 1: Frontend Developer
- Sprint 2: Product Owner and Frontend Developer
- Sprint 3: Backend Developer



Individual Reflection for Jacob



Contribution to
the product

- Sprint 1: Login/ Register UI
- Sprint 2: Route Protection, Persistent Data
- Sprint 3: Refining route protection for different user types



Individual Reflection for Jacob



Key Takeaway:
The Importance of
Granularity of tasks and
effects of Dependencies

- First sprint, tasks too generic
- Dependencies not taken into account
- Many people working on 1 generic task → Hard to track progress
- Also led to the creation to many sub tasks on the fly which affected our burndown chart → Raised it to 400% in the first sprint



Individual Reflection for Jacob



Key Takeaway:
The Importance of
Granularity of tasks and
effects of Dependencies

Resolution of this issue

- Figuring out what is needed in what task → What components are needed to fulfill a task
- Having clearer responsibilities on which person is responsible for each part of the application → For example, if someone needed help with getting some sort of persistent data, I would step in



Individual Reflection for Lucas Lonegro Gurfinkel



Roles assumed in
each sprint

- Sprint 1: Product Owner, Backend Developer
- Sprint 2: Backend Developer
- Sprint 3: Backend Developer



Individual Reflection for Lucas Lonegro Gurfinkel



Contribution to
the product

- Sprint 1: Setting up backend API on Java Spring, base models and endpoints
- Sprint 2: Refine review endpoints/search, administrator functionalities
- Sprint 3: Database and endpoints for review comments



Individual Reflection for Lucas Lonegro Gurfinkel



Key Takeaways:
Importance of
cross-functional teams
Documentation that is
not 'waste'

Initial Perspective: Frontend developers
can ask about API usage and properties

Problem: Inconsistent knowledge,
misunderstandings, unknown unknowns

Actions: Written documentation for
backend API - available endpoints and
usage - as part of DoD



Individual Reflection for David Viskovic



Roles assumed
in each sprint

- Sprint 1: Scrum Master and Frontend Developer
- Sprint 2: Frontend Developer
- Sprint 3: Frontend Developer



Individual Reflection for David Viskovic



Contribution to
the product

- Sprint 1: Login/ Register Functionality (Frontend)
- Sprint 2: Defining manual frontend testing, Refining Register Functionality
- Sprint 3: Refine website design, Refining manual frontend testing



Individual Reflection for David Viskovic



Key Takeaway:
Test Driven
Development

- unknown errors popping out
- huge time waste finding errors
- build in main branch erroneous
- importance of testing after pulling and before building
- Now: diligently adopt practice



Individual Reflection for Shan Hong



Roles assumed in
each sprint

- Sprint 1: Frontend Developer
- Sprint 2: Scrum Master and Frontend Developer
- Sprint 3: Frontend Developer



Individual Reflection for Shan Hong



Contribution to
the product

- Sprint 1: Designing the home page UI
- Sprint 2: Designing the course details page and admin page
- Sprint 3: Implementing comments and like/dislike features
- Coordinate with backend developers to fix bugs



Individual Reflection for Shan Hong



Key Takeaways

- Foresee potential challenges and plan ahead to mitigate them
- Delegate tasks based on the availability of the team during a sprint
- Reassign task based on the team members progress during stand-ups



Individual Reflection for Quang Anh



Roles assumed in
each sprint

- Sprint 1: Frontend Developer
- Sprint 2: Frontend Developer
- Sprint 3: Product Owner with Zheng Wei and Frontend Developer



Individual Reflection for Quang Anh



- Sprint 1: Course review UI
- Sprint 2: Course review functionality
- Sprint 3: Like/Dislike feature

Contribution to
the product



Individual Reflection for Quang Anh



Key Takeaways

- Clear communications regarding product goals is important.
- With additional functionalities and customer requests, prioritization is necessary to complete higher priority items first and meet sprint and product goals.



Conclusion

What Went Well

We grew into a self-organized, committed team that complemented each other's strengths during the project.

What We'd Do Differently

If we started again, we'd invest more time in planning our sprint process. But first, we had to gain experience and get familiar with the agile framework.

Did We Succeed?

Yes, we delivered our MVP and worked as an agile team.



06

Question Round





THANK YOU FOR YOUR ATTENTION!

CREDITS: This presentation template was created by [Slidesgo](#), and includes icons by [Flaticon](#), and infographics & images by [Freepik](#)

